December-23-03-5026

UG Computer Application Certificate Course Semester-I (NEP) Examination

Course Implemented From June-2023

Dec. -2023

Skill Enhancement Course

Mathematics – I – MS23SECBCA106

Total Marks : 25 **Time : 2 Hours**

Que. 1 Solve the following. (Any Two) 10

1. State and prove De Morgan's Law.
2. $1 + 3 + 5 + 7 +$ _____ $+(2n-1) = n^2$
3. If A = set of letters of word COLLEGE,

   B = set of letters of word SCHOOL

   Then find $A \Delta B$

Que. 2 Solve the following. (Any Two) 10

1. Let a and b be integers and suppose Q(a,b) is Defined recursively by
   $Q(a,b) = \{(Q(a-b, a+b) + b, a \geq b$
   Find Q(2,7), Q (5,3) and Q(15,2)
2. Let f: R-{-1}-> R, $f(x) = \frac{1-x}{1+x}$ find fof.
3. Let f: R->R and g: R->R, $f(x) = 5x^2 - 2x + 4$, $g(x) = x^3 - 2$ find the formula of fog(x) and also find Fog (-2).

Que. 3 Answer the Following. 05

1. Universal Set.
2. Power Set.
3. One-to- One Function.
4. Floor Function.
5. $\text{Log}_{10}\ 0.001$

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